

Kirloskar Chillers Private Limited

A Kirloskar Group Company

Document Name	Aluminum and Copper Cable Ratings				
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Date	06/01/2021	Prepared By	DBM	Approved By	ASP

Notes:

1. For Chillers supplied with standalone VFD, copper cables are mandatory.
2. Maximum cable length should be 30 m, if longer length is required one size higher cable should be used.
3. Current ratings are based on below considerations
 - a) XLPE insulated armoured cables are considered
 - b) Ambient temperature 50°C
 - c) Ground temperature (For cable in duct) 50°C
 - d) Part of cable laid on cable trays exposed to air
 - e) Part of cable laid in duct
 - f) Minimum clearance between wall and cable is 20mm
 - g) Cables with multiple runs are touching

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Aluminum Cable Size (Armored 3.5 Core)	No. of Runs	Max allowable Current	Tick Mark Applicable
Sq. mm	#	Amps	
50	1	101	
70	1	128	
35	2	145	
95	1	158	
120	1	182	
50	2	177	
150	1	207	
70	2	224	
185	1	240	
95	2	276	
240	1	286	
120	2	319	
70	3	305	
150	2	364	
95	3	376	
185	2	421	
120	3	433	
240	2	502	
150	3	494	
300	2	579	
185	3	572	
150	4	625	
240	3	682	
185	4	724	
300	3	786	
240	4	862	
400	3	915	
300	4	994	

Copper Cable Size (Armored 3.5 Core)	No. of Runs	Max allowable Current	Tick Mark Applicable
Sq. mm	#	Amps	
35	1	107	
50	1	131	
70	1	165	
35	2	187	
95	1	204	
120	1	234	
50	2	229	
150	1	266	
70	2	289	
185	1	308	
95	2	357	
240	1	365	
120	2	410	
70	3	393	
150	2	467	
95	3	485	
185	2	540	
120	3	557	
240	2	640	
150	3	635	
300	2	735	
185	3	734	
150	4	803	
240	3	870	
185	4	928	
300	3	999	
240	4	1100	
400	3	1152	
300	4	1263	